MIAE 221: Materials Science and Engineering

Recent Post-2020 Midterm Exam Archive & Multi-Instructor Solutions Guide (2021–2026)

Concordia University · Gina Cody School of Engineering and Computer Science

Curriculum Scope: Callister 10th/11th Edition (Chapters 1 through 7.4) & Lectures 1–12

Executive Summary & Multi-Instructor Landscape

At Concordia University, **MIAE 221 (Materials Science)** is coordinated across multiple lecture sections. While foundational course syllabi and lecture slides originate from core curriculum coordinators (including Dr. Mamoun Medraj and Dr. Martin Pugh), term midterm examinations are authored across different sections by active faculty:

- **Dr. Mamoun Medraj** (Core Coordinator & Lecture Author)
- **Dr. Martin Pugh** (Sections X, X-A, T, Y)
- **Dr. Navid Sharifi** (Sections T, Y, and Winter terms)
- **Dr. Mamadou Sy** (Winter terms)
- **Dr. Sam Eskandarian** (Section X)

Because all sections share identical accreditation criteria and test Callister Chapters 1–7.4, studying midterm examinations from across all post-2020 sections provides comprehensive mastery over every conceivable exam question archetype.

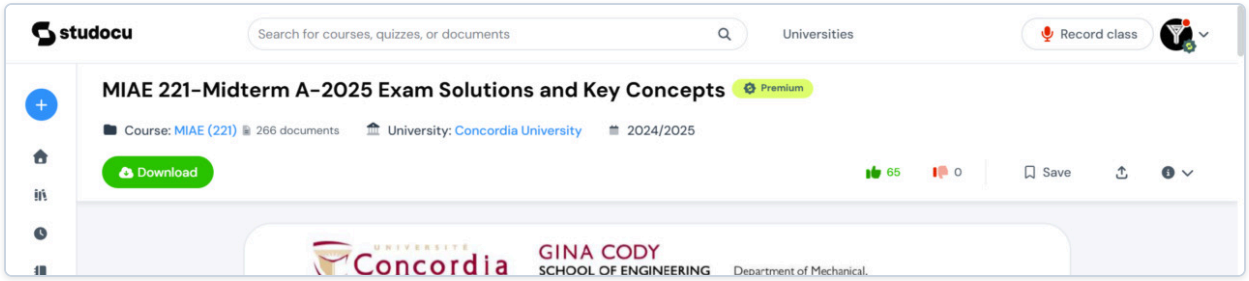
Table of Documented Recent Post-2020 Midterm Examinations

Exam Paper & Term	Instructors / Section	Document ID / Attribution	Core Tested Archetypes
2025 Midterm Exam (Version A)	General Midterm A	Studocu ID `124560047`	Interatomic potentials, FCC/BCC lattice packing, Bragg diffraction peaks, interstitial diffusion.
Winter 2026 Midterm Exam	Drs. Mamadou Sy & Navid Sharifi	Studocu ID `159540505`	Linear/planar atomic density proofs, Hume-Rothery solubility, error-function non-steady diffusion.
Fall 2025 Midterm Exam (Version A)	Section X · Dr. Sam Eskandarian	Studocu ID `144474318`	Stress-strain curve landmarks, Schmid's Law resolved shear stress, dislocation slip geometry.
Fall 2024 Midterm Exam	Sections T & Y · Drs. Pugh & Sharifi	Studocu ID `108649883`	Miller-Bravais HCP indices, vacancy Arrhenius calculations, Cu-Ni interdiffusion couples.
Fall 2024 Midterm Exam	Section X-A · Dr. Martin Pugh	Studocu ID `112451179` / `110496507`	Grain boundary energy classification, True stress vs engineering stress, Hall-Petch strengthening.

Exam Paper & Term	Instructors / Section	Document ID / Attribution	Core Tested Archetypes
2025 Structured Practice Midterm	Official Practice Review	Studocu ID `174494848`	Diagnostic self-assessment, QR-code verified solution rubrics, conceptual multiple choice.

Detailed Analysis of Recent Post-2020 Midterm Papers

1. 2025 Midterm Examination — Version A



Exam Header 1 — 2025 Midterm Version A

- **Official Header**: `MIAE 221 – Materials Science – Midterm 2025, Version A`
- **Source Reference**: Studocu Document ID `124560047`
- **Key Concepts Analyzed**:

1. **Interatomic Force vs. Energy Curves**:

- Equilibrium spacing r_0 occurs where net force $F_N = 0$, but where net potential energy E_N is at its absolute minimum trough ($\frac{dE}{dr} = 0$).
- Deep, steep potential wells correspond to high melting temperature (T_m), high elastic modulus (E), and low coefficient of thermal expansion (α).

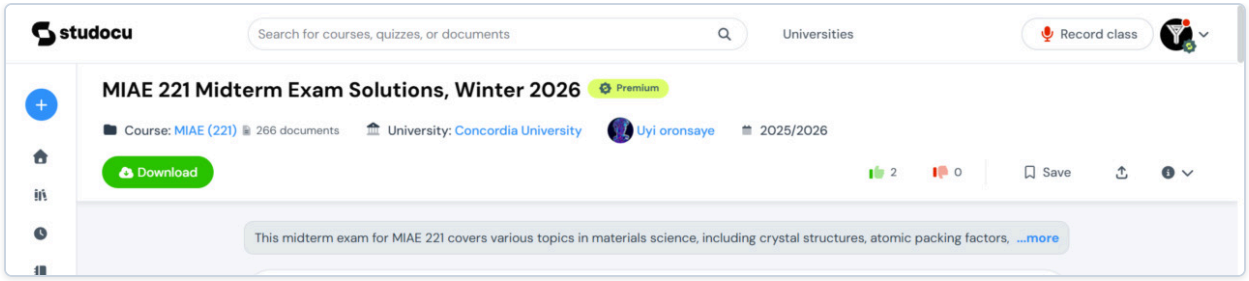
2. **FCC & BCC Geometry**:

- FCC close-packed directions: $\langle 110 \rangle$, close-packed planes: $\{111\}$.
- BCC close-packed directions: $\langle 111 \rangle$, close-packed planes: $\{110\}$.

3. **Bragg's Law and Selection Rules**:

- BCC reflects only planes where $h + k + l = \text{even}$ (first reflection: (110), second: (200)).
- FCC reflects only planes where h, k, l are all unmixed (all even or all odd: (111), (200), (220), (311)).

2. Winter 2026 Midterm Examination — Drs. Mamadou Sy & Navid Sharifi



Exam Header 6 — Winter 2026 Midterm

- **Official Header**: `MIAE 221 – Midterm, Winter 2026; Instructors: Drs. Mamadou Sy & Navid Sharifi`
- **Source Reference**: Studocu Document ID `159540505`
- **Key Concepts Analyzed**:

1. **Non-Steady-State Diffusion (Fick's Second Law)**:

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- When determining case depth or carbon penetration time, $\frac{x^2}{Dt} = \text{constant}$.
- Doubling the diffusion depth requires increasing the heat-treatment time by a factor of 4 ($t_2 = 4t_1$).

2. **Linear and Planar Density Calculations**:

- Planar Density (PD) on (110) plane in BCC iron:

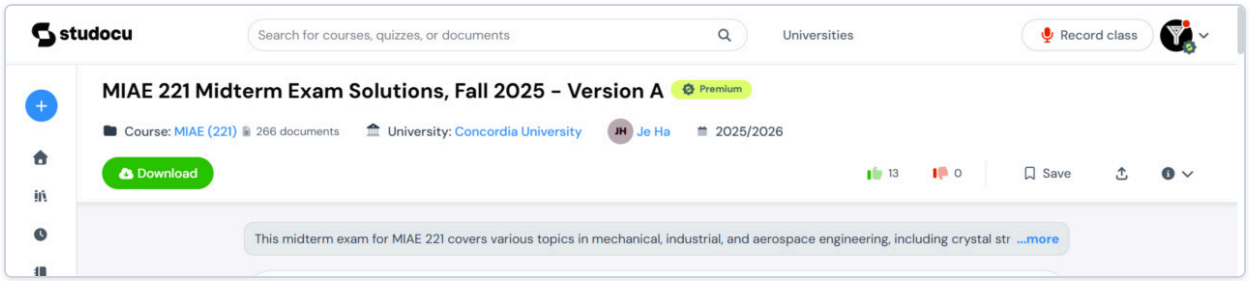
$$PD_{(110)} = \frac{\text{Number of equivalent atoms centered on plane}}{\text{Area of plane}} = \frac{2}{\sqrt{2} a^2} = \frac{\sqrt{2}}{a^2}$$

- Linear Density (LD) along $[110]$ direction in FCC copper:

$$LD_{[110]} = \frac{2 \text{ atoms}}{\sqrt{2}a} = \frac{2}{4R} = \frac{1}{2R}$$

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3. Fall 2025 Midterm Examination — Section X (Dr. Sam Eskandarian)



Exam Header 7 — Fall 2025 Midterm Version A

- **Official Header**: `MIAE 221/X – Midterm, Fall 2025; Instructor: Dr. Sam Eskandarian`
- **Source Reference**: Studocu Document ID `144474318`
- **Key Concepts Analyzed**:

1. **Schmid's Law & Critical Resolved Shear Stress (CRSS)**:

$$\tau_R = \sigma \cos \phi \cos \lambda$$

- Slip initiates when the resolved shear stress reaches the critical value: $\tau_R \ge \tau_{\text{CRSS}}$.
- The maximum resolved shear stress occurs when $\phi = \lambda = 45^\circ$, giving $\tau_R = \frac{\sigma}{2}$ (Schmid factor $m = 0.5$).
- If slip direction or slip plane normal is perpendicular to the tensile axis ($\phi = 90^\circ$ or $\lambda = 90^\circ$), $\tau_R = 0$ and no slip occurs.

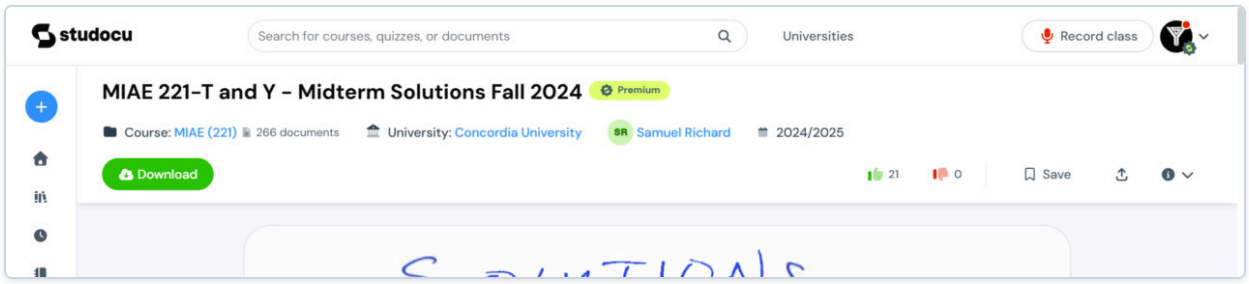
2. **Engineering vs. True Stress/Strain**:

$$\sigma_T = \sigma_E(1 + \epsilon_E) \quad \text{and} \quad \epsilon_T = \ln(1 + \epsilon_E)$$

(Valid up to the onset of necking at maximum load / UTS).

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4. Fall 2024 Midterm Examination — Sections T & Y (Drs. Martin Pugh & Navid Sharifi)



Exam Header 5 — Fall 2024 Sections T & Y

- **Official Header**: `MIAE 221 – T and Y – Midterm Solutions Fall 2024; Instructors: Drs. Martin Pugh & Navid Sharifi`
- **Source Reference**: Studocu Document ID `108649883`
- **Key Concepts Analyzed**:

1. **Point Defects & Arrhenius Vacancy Concentrations**:

$$N_v = N \exp\left(-\frac{Q_v}{k T}\right)$$

- $N = \frac{\rho N_A}{A}$ (atomic sites per unit volume).
- Temperature T must always be converted to Kelvin ($K = ^\circ\text{C} + 273.15$).
- Boltzmann constant $k = 8.617 \times 10^{-5}\text{ eV/K}$ or $1.38 \times 10^{-23}\text{ J/K}$.

2. **Hume-Rothery Rules for Complete Solid Solubility**:

- $\Delta R \le 15\%$ (atomic size factor).
- Crystal structures must be identical (e.g. both FCC for Cu and Ni).
- Electronegativities must be similar.
- Valences must be identical or higher valency metal preferred as solute.

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5. Fall 2024 Midterm Examination — Section X-A (Dr. Martin Pugh)



Exam Header 2 — Fall 2024 Section X-A

- **Official Header**: `MIAE 221 – Materials Science – Midterm Fall 2024, Section X-A; Instructor: Dr. Martin Pugh`
- **Source Reference**: Studocu Document ID `112451179` & `110496507`
- **Key Concepts Analyzed**:

1. **Diffusion Mechanisms**:

- Vacancy (substitutional) diffusion requires both vacancy creation and atomic jumps ($Q_d = Q_v + Q_m$).
- Interstitial diffusion (C, H, N, O in iron) occurs much faster with lower activation energy because interstitial sites are largely empty.

2. **Stress-Strain Landmarks**:

- Proportional Limit: end of strictly linear elastic behavior.
- 0.2% Offset Yield Strength: draw line parallel to modulus slope at $\epsilon = 0.002$.
- Ultimate Tensile Strength (UTS): maximum engineering stress prior to necking.
- Fracture Stress: stress at complete separation (lower than UTS in engineering stress, but highest in true stress).

Core Formula Reference & Exam Quick-Check

1. **Lattice Dimensions & Atomic Packing**:

- FCC: $a = 2\sqrt{2} R$, Coordination Number $= 12$, Atoms/cell $n = 4$, $\text{APF} = 0.74$.
- BCC: $a = \frac{4}{\sqrt{3}} R$, Coordination Number $= 8$, Atoms/cell $n = 2$, $\text{APF} = 0.68$.
- HCP: $c/a = 1.633$, Coordination Number $= 12$, Atoms/cell $n = 6$, $\text{APF} = 0.74$.

2. **Interplanar Spacing (Cubic)**:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

3. **Bragg's Diffraction Law**:

$$\lambda = 2 d_{hkl} \sin \theta$$

4. **Temperature-Dependent Diffusion Coefficient**:

$$D = D_0 \exp\left(-\frac{Q_d}{R T}\right)$$

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